

Research Documentation: AGB Mapping in the Rwenzori Mountains

Methodology: Multi-Sensor Fusion & Deep Learning ("Flexible Net")

1. Data Engineering Philosophy (GEE)

The transition from traditional Forest Canopy Density (FCD) models to an **NDVI-based Radar Fusion** was a strategic pivot necessitated by the environmental constraints of the Rwenzori Mountains.

1.1 Transition from Traditional FCD (AVI/BI/SI) to NDVI-Fusion

While AVI, BI, and SI are standard for optical forest monitoring, they were replaced in this study for three specific reasons:

- **Multi-Modal "Cloud-Proofing"**: Traditional indices fail simultaneously under cloud cover. By using NDVI, we leverage its well-documented linear relationship with **Sentinel-1 VH Backscatter**. This allows the "Flexible Net" to "fill the gaps" using radar when optical data is obscured.
- **Mitigating Saturation**: In high-biomass tropical canopies, indices like AVI "blow out" (saturate) quickly. NDVI, while also prone to saturation, is more robust when paired with the **Canopy Height Model (CHM)**, providing two distinct dimensions (Horizontal Greenness + Vertical Structure) rather than overlapping optical indices.
- **Orthogonal Feature Engineering**: Neural networks converge faster when inputs represent different physical properties. Our chosen stack minimizes redundancy:
 - **NDVI**: Photosynthetic health (Leaves).
 - **CHM**: Vertical structure (Volume).
 - **CDM**: Fused horizontal cover (Density).
 - **VV**: Surface roughness (Texture/Stems).

1.2 Sentinel-2 Pre-processing Improvements

- **SCL-Based Masking**: We utilized the **Scene Classification Layer (SCL)** instead of the standard QA60 bitmask. SCL is superior for the Rwenzori terrain because it explicitly identifies **Class 3 (Cloud Shadows)** and **Class 11 (Snow/Ice)**, which are frequently misclassified in rugged, high-altitude regions.
- **Harmonized Collection**: By using `COPERNICUS/S2_SR_HARMONIZED`, we ensured that the 2023 data remains radiometrically consistent with earlier processing baselines, preventing artificial "drift" in NDVI values.

2. Pre-Inference Validation: The CDM Protocol

Before utilizing the full stack for AGB estimation, the **Canopy Density Model (CDM)** must be validated to ensure the Radar-Optical fusion accurately represents the Rwenzori's canopy.

2.1 Validation Methods

1. **Qualitative (Visual High-Res):** Overlaying CDM on Google Satellite basemaps to ensure clearing/gaps in the imagery match low-CDM pixels.
2. **Quantitative (GEDI LiDAR):** The "Gold Standard." Comparing 10m CDM pixels against **NASA GEDI (L2A/L2B)** 25m circular footprints for `canopy_fraction`. A Pearson correlation (r) of >0.70 is the target for a reliable structural model.
3. **Cross-Dataset Comparison:** Using a **Difference Map** against the **Hansen Global Forest Change** dataset.
 - **Blue Areas:** CDM > Hansen (Radar capturing sub-canopy density).
 - **Red Areas:** CDM < Hansen (Potential over-masking of Sentinel-2 pixels).

2.2 Statistical Reporting

The research will report **RMSE**, R^2 , and **Bias** for the CDM to justify its use as a primary input for the Flexible Net.

3. Training & Inference Workflow (Python/Colab)

The project follows a strict **Two-Phase** execution:

Phase 1: The Training Phase (`training.py`)

- **Allometric Grounding:** Individual tree DBH and Height are converted to plot-level AGB using the **Chave et al. (2014)** equation.
- **Point-to-Pixel Extraction:** Pixel values from the `finalStack` are extracted at plot GPS coordinates to create the training table.
- **Weight Stabilization:** The model utilizes **Batch Normalization** to reconcile different sensor scales and **Dropout (0.2)** to ensure the network generalizes across the park rather than overfitting to specific plots.

Phase 2: The Inference Phase (`flexiblenet_inference.py`)

- **Intelligence Transfer:** The trained weights (`.h5` file) are loaded into a fresh instance of the Flexible Net.
- **Batch Processing:** To circumvent Google Colab RAM limits, the Rwenzori GeoTIFF is processed in **500-row windows**, outputting a single-band 10m resolution AGB map that preserves the original **UTM Zone 35N** spatial metadata.

Comparison of Methodologies

Feature	Traditional FCD	Rwenzori Flexible Net
Cloud Resistance	Low (Optical Only)	High (S1 + S2 Fusion)
Data Redundancy	High (Correlated Bands)	Low (Orthogonal Features)
Accuracy Metric	Spectral Correlation	Physically-grounded AGB